Work, Power & Energy

Work :

(i) $\boxed{W = \vec{F}\cdot\vec{d}}$ [d → displacement of point of action of force in dirⁿ of force]

(ii) Units : Joule (J) → SI unit
Erg → CGS unit

(iii) +ve work → Energy supplied
−ve work → Energy consumed

(iv) For any centripetal force, $\boxed{W_F = 0}$ always

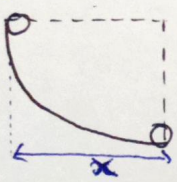
Imp
(v) [Work done by a force on a body] ⇒ [Frame Dependent]
[Net work done by the force] ⇒ [Frame Independent]

Work done by friction :

$\boxed{W_{kinetic\,fr} = -ve}$
$\boxed{W_{static\,fr.} = 0}$ } always

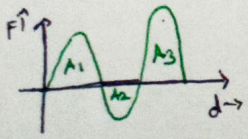
→ If path is ~~tangen~~ curved & friction always acts tangential, then

$\boxed{W_f = -\mu mg X}$



Graph :

Area under F−d curve gives work done by F.



$\boxed{W_F = A_1 - A_2 + A_3}$

$\boxed{\int dW = F\cdot\int dr}$

Work Done by Variable forces :→

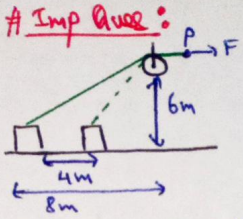
(i) 1-Dimensional ⇒

$$\boxed{W = \int_{r_1}^{r_2} f(r)\cdot dr}$$

(ii) 2-Dimensional ⇒

$$\boxed{W = \int_{x_1}^{x_2} F_x\cdot dx + \int_{y_1}^{y_2} F_y\cdot dy}$$

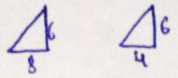
Imp Ques :



Jitni rassi neeche se free hui, utna hi displacement point of action (P) Ka hua

$\Delta l = 10 - \sqrt{52}$

$\boxed{W_F = F(10 - \sqrt{52})}$



Energy :

Good / Organised	Bad / Unorganised
Easily convertible (KE, PE)	Conversion very unfeasible (Heat, Sound)

Kinetic Energy :

$$\boxed{KE = \frac{1}{2}mv^2 = \frac{p^2}{2m}}$$ (p → Linear Momentum)

Work energy Theorem ⇒

$\boxed{W_{net} = \Delta KE}$

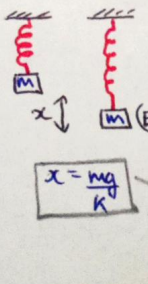
Work Done by spring ⇒

$$\boxed{W_{spring} = -\frac{1}{2}k\left[(x_f)^2 - (x_i)^2\right]}$$

[x measured from mean position]

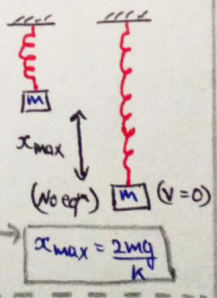
Max. elongation in spring ⇒

(a → 0)



$\boxed{x = \frac{mg}{k}}$

(a ≠ 0)



Do guna Lagaan!

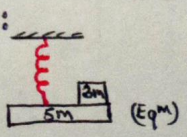
$\boxed{x_{max} = \frac{2mg}{k}}$

Also, (V. Imp)

$\boxed{x_{max} = \frac{2\cdot F_{sudden}}{k}}$

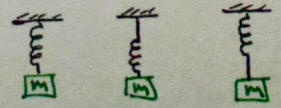
[3m block suddenly placed at eqm]

Eg:



$x_{max} = \frac{2(3mg)}{k}$

→ NOTE :→



If string & spring, both added

x_{max} ⇓ $\left(\frac{2mg}{k}\right)$ ~~[illegible]~~ in all

Force as f(t) :

$$\int F\cdot dt = m\int dv$$

$\boxed{\text{Area under F-t curve} = \text{Change in momentum } (\Delta p)}$

Conservative Forces :

- → Convert good form into good form only
- → Const. forces / Forces with central nature
- → Gravitational / Electrostatic / Spring

Properties ⇒

(i) W independent of path, depend only on x_i & x_f

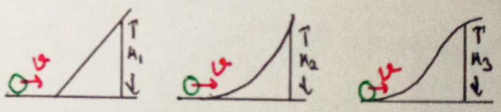
(ii) W in closed loop is zero

(iii) Condⁿ to be Conservative ⇒

$$\vec{\nabla}\times\vec{F} = 0$$

$$\boxed{\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)\times\vec{F} = 0}$$

$$\boxed{\left(\frac{\partial F_y}{\partial x} - \frac{\partial F_x}{\partial y}\right)\vec{k} = 0}$$ {For 2-D forces}



∴ mg ⇒ conservational force

$\boxed{H_1 = H_2 = H_3}$

→ [Friction / Air drag / Viscous] [Non Conservative]

Potential Energy :

↳ It is −ve of work done by Conservative force.

$$\boxed{\Delta U = -F_{cons}\cdot\int_{r_1}^{r_2} dr}$$

G.P.E ⇒ $\boxed{\Delta U = mgH \,/\, -mgH}$

Elastic PE ⇒ $\Delta U = -W_{spring}$
$= \frac{1}{2}k\left[(x_f)^2 - (x_i)^2\right]$

Conservation of Mechanical energy ⇒

(i) All non-conservative force absent

(ii) Inertial frame observer (Frame dependent force = 0)

Then $\boxed{\Delta KE + \Delta PE = 0}$

→ PE is relative quantity & defined w.r.t zero potential line

→ Generally, ground is Z.P.L

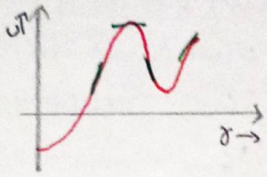
Imp → PE → Depends on Zero Pot. Line
ΔPE → Independent of zero. Pot. Line

Potential Energy Curve ⇒

∴ $dU = -F \cdot dr$

$$F = -\left(\frac{dU}{dr}\right)$$

-ve of slope of PE curve gives force



→ Maxima ⇒ Unstable eqm

→ Minima ⇒ Stable eqm

→ If particle is moving in space & PE is variable, then

$$\vec{F} = -\left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right)$$

Power :

Avg. Power ⇒ $\frac{\text{Work Done}}{\text{Time Taken}} = \frac{W}{\Delta t}$

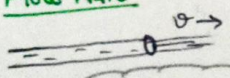
Inst. Power ⇒ $\frac{dW}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$

Unit : SI Unit ⇒ 1 Watt = 1 J/s
other Unit ⇒ 1 HP = 746 W

Gear Box ⇒

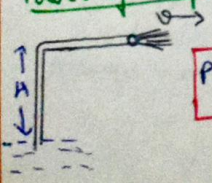
At lower Gear ⇒ F↑, v↓
At High Gear ⇒ F↓, v↑
} Overall $\vec{F} \cdot \vec{v}$ = const.

Flow Rate ⇒



Volume per sec = $A \cdot v$
Mass per sec = $\rho \cdot A \cdot v$

Power of Pump ⇒



$$P = \frac{1}{2}\rho A v^3 + (\rho A v) g H$$

→ If height is negligible, then

$$P = \frac{1}{2}\rho A v^3$$

⇒ Power ∝ (velocity)^3

Dependency of Power ⇒

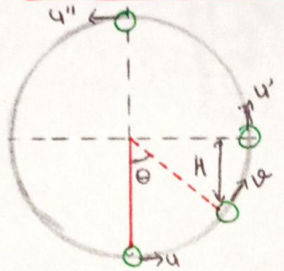
Power

$$P = m \cdot \frac{dv}{dt} \cdot v \qquad P = m \cdot v^2 \cdot \frac{dv}{dx}$$

(i) KE ∝ t ⇒ $v \propto \sqrt{t}$

(ii) $v \propto (x)^{1/3}$

Vertical Circular Motion :



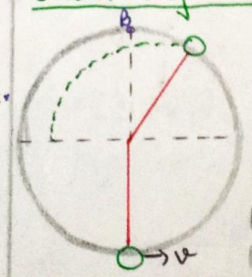
→ Bob suspended by ideal string

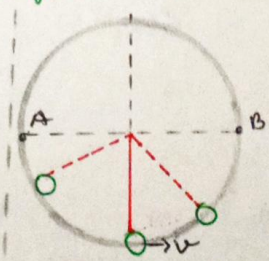
→ Body moving in a hollow circle

1) $u = \sqrt{5gR}$
$u' = \sqrt{3gR}$
$u'' = \sqrt{gR}$
} Difference of $\sqrt{2gR}$ after each quarter.

2) $v = \sqrt{u^2 - 2gH}$

Condition of leaving circular path ⇒





$\sqrt{2gR} < v < \sqrt{5gR}$
Body starts projectile before reaching B.

$v \leq \sqrt{2gR}$
Body oscillates b/w A and B

Point of leaving ⇒

$$\cos\theta = \frac{u^2 - 2gR}{3gR}$$

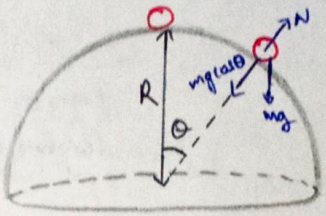
Case of light Rod ⇒

→ Since rod cannot slack & can apply outward N force unlike string.

$$u_{min} = \sqrt{4gR}$$

Other things are same as string-bob system.

Cond^n of leaving hemisphere ⇒



$$\theta = \cos^{-1}\left(\frac{2}{3}\right)$$